# Smart School Food Service Analysis: AI-Driven Demand Forecasting and Waste Reduction Using Fairfax County Public Schools Data

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*Abstract*—**Food waste in U.S. public schools is a major economic and environmental problem, with about 530,000 tons of food discarded each year and $1.7 billion in losses [1]. Fairfax County Public Schools (FCPS) in Virginia plans to solve this problem by using data to reduce waste and improve efficiency. While digital data from Point-of-Sale (POS) systems and kitchen records is widely available, most districts, including FCPS, lack the tools to transform this data into actionable insights. This study analyzes FCPS data from May 2025 to pinpoint inefficiencies, identify high- and low-demand items, and reveal patterns by region and school level. Our main goal is to provide clear, practical recommendations that allow FCPS to serve meals more sustainably and efficiently.**

## I. INTRODUCTION

School food service operations face mounting pressure to deliver nutritious meals while minimizing waste and controlling costs. As student populations grow more diverse and operational demands more complex, school systems urgently need intelligent, data-driven solutions to optimize meal planning, procurement, and production.

Beyond inefficiencies, school meals play a critical role in food security and diet quality. For many children living in low income or food insecure households, school breakfasts and lunches represent a major portion of their daily caloric intake, yet these meals consistently outperform non-school foods in diet quality. In one national sample, USDA served foods scored between 79–81 on the Healthy Eating Index versus 55–57 for non-school foods [2]. This underscores that food service operations aren't just about cost savings and that improving their efficiency also directly supports children's health and equity.

At the same time, data suggest that recent policy driven improvements in meal standards have boosted consumption without increasing waste. A study comparing plate waste before and after the 2012 USDA school meal standard update found that students consumed 15.6% more of their entrée and 16.2% more vegetables after the new guidelines took effect [3]. These results contradict narrative driven claims that stricter nutrition rules necessarily lead to more food being thrown away and point instead to a compelling case for data driven planning and production.

Even with these gains, most school districts lack sophisticated analytics tools able to harness point-of-sale (POS) and production data in real time. Without such capabilities, opportunities to further reduce waste, align procurement, and tailor menus remain largely untapped. To take initiative in solving this problem, Fairfax County Public Schools (FCPS) has provided data for March to May of 2025, offering a timely opportunity to develop and deploy advanced, data-driven solutions.

## II. RELATED WORK

### A. The Cafeteria as a Classroom

In January 2019, Gerin Hennebaul, a teacher at Lovin Elementary School in Lawrenceville, Georgia, and a group of students sorted leftover milk, fruit, vegetables, and other foods into buckets during "Taco Day" [4]. This specific day, 721 children were served and as a result 600 pounds of waste was generated, where 75 pounds were fruit and vegetable waste, and 120 pounds of uneaten food [4]. The amount of food of this magnitude being transported into a landfill will cause environmental problems as the waste will emit methane into the environment. To address this, the World Wildlife Fund works with about 250 schools throughout the United States to teach the children about food waste and climate change. On the federal level, the USDA, who is responsible for feeding the students in schools, has not made the effort to establish reducing food waste as a core part of school nutritional programs [4]. This makes it difficult for school districts to find ways to reduce food waste as they would have to find their own ways to keep waste low.

For their efforts, Lovin Elementary made a number of changes to address their food waste problems. First, the school made an announcement during their morning announcements to state that milk is not necessary for students to take for their lunch. Students felt prior that milk was mandatory and if they did not want to drink the milk, then they would leave it wasted [4]. The leftovers from the teachers' salad bar was feed to the school's chickens and a shared table was created for unopened food and fruit where students can take and leave food for others [4]. The school has also increased their composting efforts, as of 2019, 5000 pounds of food was composted to the school's garden and local organizations [4]. In March of the same year, food waste had dropped from 589 to 435 pounds, showing promising results [4].

### B. Study of Waste Production at a Private School

Building on these approaches, other research teams also studied which foods were most wasted to better target solutions.

In [5], a study by Christine Costello, assistant professor at Penn State, focused on a private school in Columbia, Missouri. The school was chosen because its administrators wanted to improve cafeteria sustainability, and it includes an unusual mix of students from ages four to eighteen [5]. Another study found that the students from all grades wasted vegetables and fruit the most as it accounted for 50% of their plates and 43% of faculty waste [5]. A particular problem is that, "there are certain foods cafeteria managers are buying, presumably to meet U.S. Department of Agriculture guidelines, that the kids never are going to eat" [5].

The study that Costello was intended to help cafeteria management and school lunch programs to understand their food production systems is causing problematic food waste that is harming the environment. Costello said, "In a detailed food-waste study like this, we can report back to menu planners what we learn about what children don't eat in order to seek menus that they will eat" [5]. Eliminating the production of unpopular food will help reduce the food waste generated, and focus on foods that students will actually eat.

### *C. Factors Influencing Food Waste*

A study conducted by the authors, Martins, Rodrigues, Cunha, and Rocha, explored the factors that contributes to food waste in 21 different primary schools in Porto, Portugal [6]. Food waste was measured by the weight, menu offering, and were also students were surveyed related to their preferences and dietary patterns [6].

## III. MIXED-INTEGER LINEAR PROGRAMMING OPTIMIZATION

Mixed-Integer Linear Programming (MILP) formulates problems into a linear objective function where, in our case, a variable takes an integer value to address resource allocation. In [7], MILP can be formulated to the following matrix notation below:

$$\begin{aligned} \min_{\boldsymbol{x},\boldsymbol{y}} \quad & \boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{d}^{\mathrm{T}}\boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \le \boldsymbol{b} \\ & \boldsymbol{E}\boldsymbol{y} \le \boldsymbol{f} \\ & \boldsymbol{A} \in \mathbb{R}^{m\times n}, \boldsymbol{E} \in \mathbb{R}^{p\times q}, \\ & \boldsymbol{b} \in \mathbb{R}^{m}, \boldsymbol{f} \in \mathbb{R}^{p}, \boldsymbol{c} \in \mathbb{R}^{n}, \boldsymbol{d} \in \mathbb{R}^{q} \\ & \boldsymbol{x} \in \mathbb{Z}^{n}, \boldsymbol{y} \in \mathbb{R}^{q} \end{aligned}$$

Which the problem instance is represented by matrices and vectors $\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{b}, \boldsymbol{f}, \boldsymbol{c}, \boldsymbol{d}$ from various sets of real numbers and integer numbers [7].

- $\min_{\boldsymbol{x},\boldsymbol{y}}$: Indicates the objective function, in which the goal is to minimize the result by choosing the optimal values for the decision variable vectors $x$ and $y$.
- $\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x}$: The dot product of cost vector $c$ and variable vector $x$, where it represents the total cost associated with the integer variables.
- $\boldsymbol{d}^{\mathrm{T}}\boldsymbol{y}$: The total cost associated with the continuous variables $y$.
- $\boldsymbol{A}\boldsymbol{x} \le \boldsymbol{b}$: The coefficient matrix $A$ for the amount of variables $x$ consumes where it has to be less than or equal to the available capacity vector $b$.
- $\boldsymbol{E}\boldsymbol{y} \le \boldsymbol{f}$: The coefficient matrix $E$ for the amount of variables $y$ consumes where it has to be less than or equal to the limit vector $f$.
- $\boldsymbol{A} \in \mathbb{R}^{m\times n}$: Represents that matrix $\boldsymbol{A}$ consists of real numbers that has $m$ constraints on $n$ integer variables.
- $\boldsymbol{E} \in \mathbb{R}^{p\times q}$: Represents that matrix $\boldsymbol{E}$ consists of real numbers that has $p$ constraints on $q$ integer variables.
- $\boldsymbol{b} \in \mathbb{R}^{m}, \boldsymbol{f} \in \mathbb{R}^{p}, \boldsymbol{c} \in \mathbb{R}^{n}, \boldsymbol{d} \in \mathbb{R}^{q}$: Limit vectors $b$ and $f$ must have $m$ and $p$ real numbers respectively. Cost vectors $c$ and $d$ must have $n$ and $q$ real numbers respectively.
- $\boldsymbol{x} \in \mathbb{Z}^{n}$: Forces every value in the vector $x$ to be an integer number.
- $\boldsymbol{y} \in \mathbb{R}^{q}$: The values in vector $y$ can be real numbers.

With the emergence of AI techniques to help businesses and other entities to solve their problems, it should not be considered when it comes to interpretability. MILPs have an advantage over these AI techniques as they are easier to understand and pinpoint which constraints of the MILP model has led to the optimal solution [8]. According to Clautiaux and Ljubić, interpretability will be a requirement when it comes to decision-making tools, as machine learning algorithms suffer in transparency and providing explanations for their results, MILPs will be useful for decision makers [8].

For our MILP model, we used SciPy's MILP function, where in the SciPy library, the function is a wrapper for the HiGHS solver [9]. Going deeper, the HiGHS Mixed Integer Programming solver is solved by a Branch-and-Cut method [9]. In [10], Genova describes the Branch-and-Cut algorithm, as a powerful algorithm as it combines the Cutting Planes and Branch-and-Bound algorithms into one.

To better understand the Branch-and-Cut algorithm, fig. 1 provides a small representation of the algorithm.

- **x- and y-axis**: represents the quantities of a menu item, determining how many to items to produce.
- **Green shaded area**: represents the intersections that fits within the budget constraints. Where if a point is inside the green area, it is within budget, but if the point is outside then it either out of our budget or out of the production range. Edges are the production bounds constraints.
- **Blue points**: represent the physically possible quantities, where each point is in a whole number as needed for a ILP model.
- **Red X**: represents where a standard linear programming model would determine as the best corner of the green area. However, this fails as it falls on a fractional value, therefore this point can not be reached.
- **Blue dashed line**: represents the cut, where the solver sees that the Red X is unachievable, so it cuts to a possible solution without losing valid points.
- **Red star**: once the cut occurs, the star represents the best integer solution for the problem.

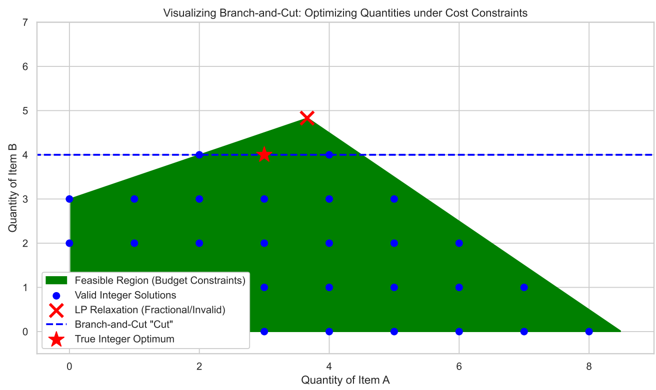


Fig. 1. Visualization of the Branch-and-Cut algorithm

*A. Optimization Model Formulation*

We created a MILP model to help FCPS cafeteria managers reduce food waste by selecting the optimal number of menu items. The model aims to reduce total costs by balancing production expenses with a waste penalty that aligns with FCPS needs.

*1) Decision Variables:* Let $x_j$ be the integer quantity of menu item $j$ to be produced.

*2) Parameters:*

- $c_j$: The unit production cost for item $j$.
- $w_j$: The waste penalty associated with item $j$ (derived from historical leftover rates).
- $d_j$: The predicted demand for item $j$.

*3) Objective Function:* The objective is to minimize the Total Cost ($C_T$), balancing the financial cost of meals and the environmental cost of waste. Based on our "Minimized Cost" metric, the objective function is:

$$\text{Minimize } Z = \sum_{j=1}^{n} (c_j + \lambda w_j) x_j \tag{1}$$

where $\lambda$ represents the waste penalty weight (set to 0.1 in our analysis).

*4) Constraints:* To align production with student needs and minimize waste, we used a production bounds constraint. This constraint keeps production within a defined range. The range is based on estimated demand:

$$L_{s,j} \leq x_{s,j} \leq U_{s,j} \quad \forall s, \forall j \tag{2}$$

In our optimization model, we scale the bounds by size multipliers tied to each school's student population (Table I). We see that smaller schools have a longer range than the larger schools. Schools with smaller populations are more volatile as a small change in daily events can result in a massive waste percentage impact. For example, a school of 500 students one day has a flu outbreak and 50 students call in sick, that is a 10% reduction in demand for that day. A wider buffer takes account for this unpredictability. On the other hand, a school of 3000 students that has 50 students call out one day would result in a 1.6% reduction in demand. The buffer would tighter as attendance is more stable. This is represented by:

TABLE I
PRODUCTION BOUNDS BASED ON SCHOOL SIZE

| **Student Population** | **Production Bounds** $(L, U)$ |
|---|---|
| 0–499 | (0.80, 1.20) |
| 500–999 | (0.85, 1.15) |
| 1000–1499 | (0.90, 1.10) |
| 1500–1999 | (0.95, 1.05) |
| 2000–2499 | (0.96, 1.04) |
| 2500–2999 | (0.97, 1.03) |
| 3000–3499 | (0.98, 1.02) |
| 3500+ | (0.99, 1.01) |

$$\begin{aligned} L_{s,j} &= \alpha_s \cdot d_{s,j} \\ U_{s,j} &= \beta_s \cdot d_{s,j} \end{aligned} \tag{3}$$

Where coefficients $(\alpha_s, \beta_s)$ are determined by school size:

$$(\alpha_s, \beta_s) = \begin{cases} (0.80, 1.20) & \text{if } s \in \text{0-499} \\ (0.85, 1.15) & \text{if } s \in \text{500-999} \\ (0.90, 1.10) & \text{if } s \in \text{1000-1499} \\ (0.95, 1.05) & \text{if } s \in \text{1500-1999} \\ (0.96, 1.04) & \text{if } s \in \text{2000-2499} \\ (0.97, 1.03) & \text{if } s \in \text{2500-2999} \\ (0.98, 1.02) & \text{if } s \in \text{3000-3499} \\ (0.99, 1.01) & \text{if } s \in \text{3500+} \end{cases}$$

Here, $d_{s,j}$ represents the demand for item $j$ at school $s$. In addition to these production bounds, we require that the production quantities, represented by the decision variables, be non-negative integers, enforced by:

$$x_{s,j} \in \mathbb{Z}^+ \quad \forall s, \forall j \tag{4}$$

Finally, we included two budget constraints in the model: a per-school budget constraint ($B_s$), and a system-wide total budget constraint ($B_{total}$):

$$\sum_j (c_j \cdot x_{s,j}) \leq B_s \tag{5}$$

- $c_j$: The unit production cost for item $j$.
- $x_{s,j}$: Quantity of items produced by school $s$.
- $B_s$: The maximum budget allocated to school $s$.

The per-school budget constraint ensures equity across all the schools in the system. It prevents larger schools from "hoarding" the budget, allowing every school to have enough funds to meet its specific demand.

$$\sum_s \sum_j (c_j \cdot x_{s,j}) \leq B_{total} \tag{6}$$

The system-wide budget constraint ensures sustainability. It ensures that the spending across the county does not exceed the allocated \$139,144,760 budget [11].

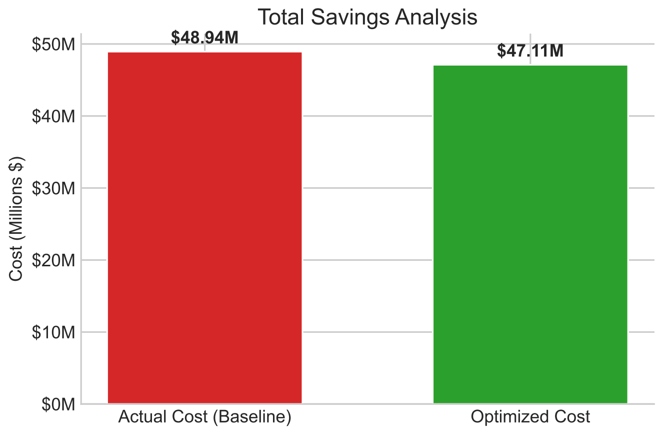


Fig. 2. Overall Savings Analysis. This comparison shows the financial impact of the optimization model across the system. The dark gray bar shows the estimated actual annual baseline cost ($48.94M), and the light gray bar shows the lower cost reached with the optimized menu planning model ($47.11M).

## IV. RESULTS

### A. Financial Impact Analysis

Fig. 2 shows a comparison of production costs between actual realized costs from the data and our model's estimated annual cost. We found that the actual annual cost for the entire school system in May 2025 was $48.94 million. After adjusting production and budgetary limits, our model estimated an annual cost of $47.11 million. This means a total savings of about $1.83 million. These results show that FCPS can achieve significant cost savings without lowering nutritional standards or reducing service.

According to the National Center for Educational Statistics [12], as of 2022, Fairfax County Public Schools:

- **Total Population**: 1,145,354
- **Median Household Income**: $145,165
- **Total Households**: 410,844
- **Race/Ethnicity**: 49% White, 20% Asian, 17% Hispanic/Latino, 10% African American, 5% Other
- **Households with Internet**: 95.9%

In terms of the FCPS regions, here are key points to describe the savings amounts from fig. 3:

- **Region 1**: The two Census districts of Dranesville and Hunter Mill make up this region and has a median household income of $199,209 [13] and $164,050 [14] for 2023 respectively. This region is the wealthiest among the county as cities like Great Falls, McLean, Wolf Trap, Vienna, and Dunn Loring are located in. Therefore, students in this region have a choice of bring in lunch or food, as their parents can afford food, resulting in adequate production costs for those that are in the lower parts of the region.
- **Region 2**:
- **Region 3**: According to Fairfax County's Needs Assessment of 2022 [15], 11 out of the 18 Census tracts in Fairfax County ranked as the greatest need for food reside in this region. This would mean that the students in these schools deal with food insecurity and benefits from the food provided by the schools. Therefore, savings in this area is low as the food production needed satisfies the demand in this region.
- **Region 4**: In [16], this region has the lowest amount of savings due to the home of the largest schools in the state, Lake Braddock Secondary, Robinson Secondary. The student population of Lake Braddock Secondary is 5422 students and Robinson Secondary is 4904 students [17]. These schools have massive student populations and so they serve far more meals than smaller high schools in other regions. Therefore, this region produces the highest volume of waste because of student population.
- **Region 5**:

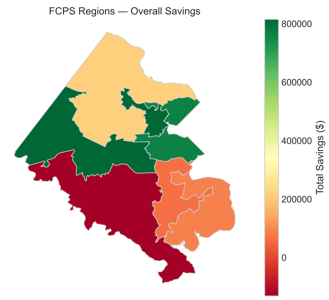


Fig. 3. Choropleth map of Fairfax County with FCPS regions divided and colored based on savings amount from optimization

Fig. 4 shows the financial performance of the optimization model. The scatter plot compares each school's actual annual food costs (x-axis) with its optimized annual costs (y-axis). The dashed diagonal line marks the point where optimized costs equal historical spending.

Schools below the line, shown in gray, are where the model found inefficiencies and projected cost savings. The model placed 130 schools in this group. In contrast, 60 schools above the line, shown in black, are expected to have higher costs. This likely happens when past procurement did not meet the model's stricter rules. The size of each marker shows the financial impact.

There is a clear correlation between school size and optimization results. The model saves the most money in schools with 500 to 1,500 students. Schools with more than 2,000 students resulted in higher costs and negative savings (see Fig. 5). Breaking down the financial data, smaller and medium-sized schools used to overproduce, which the model fixed. In contrast, larger schools may not have produced enough for their students. The model increased food allocations for these schools to ensure students are properly served.

### B. Interpretation of Projected Cost Increases

The optimization model projected a cost increase for a small group of schools, as indicated by data points above the break-even line in Fig. 4. While reverting these schools to historical production patterns may appear financially prudent, this perspective fails to account for the model's embedded constraint logic.

The optimization strictly enforces demand satisfaction constraints ($x_{s,j} \geq L_{s,j}$). This ensures production matches estimated

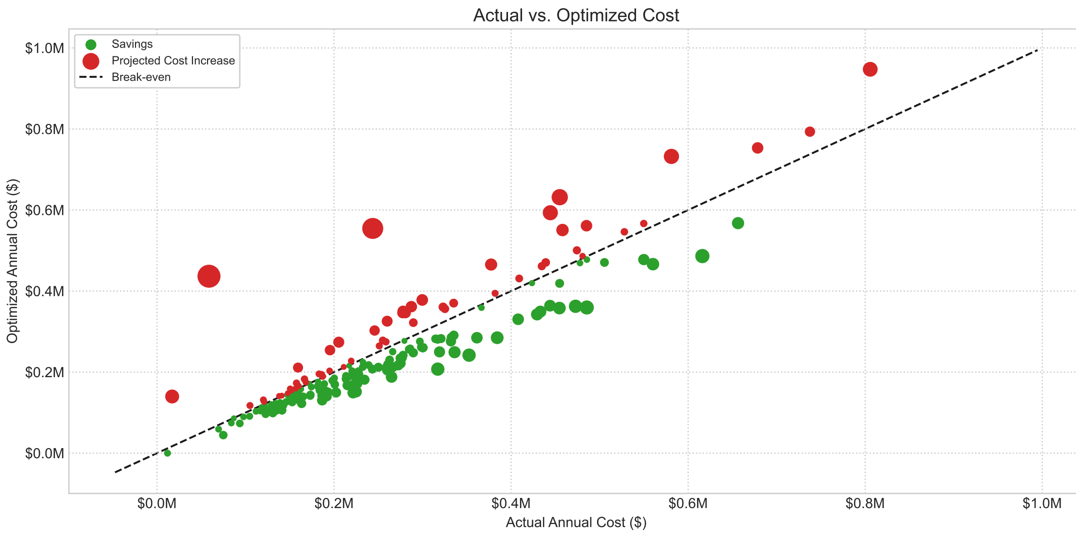


Fig. 4. Savings Analysis: Actual vs. Optimized Cost. Gray circles represent schools achieving savings, while black circles indicate projected losses. The size of the circle corresponds to the magnitude of the financial impact.

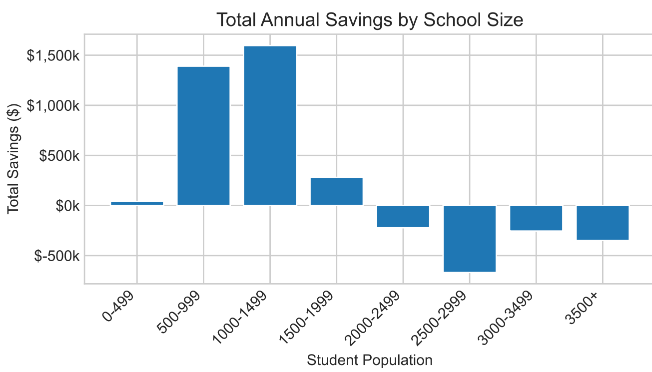


Fig. 5. Total Annual Savings by School Size Category. The chart shows that, by student population, the optimization model usually leads to net savings (positive values), but in some cases results in higher costs when demand constraints must be met (negative values).

student consumption. The historical ”savings” in these schools likely result from under-production, when supply did not meet demand. This may have led to items running out or reduced service levels. The model’s projected cost increase should not be seen as inefficiency, but as the *true cost of service assurance*. For these schools, the model shows a gap between past budgeting and the real resources needed to fully serve students.

## V. Challenges and Limitations

### A. Challenges

An issue that other researchers observed was student acceptance of certain foods. In Mulhollem’s article [5], they observed that students, particularly elementary school students, avoided certain foods that were served to them. “Because they are the most highly wasted, understanding the extent to which fruit and vegetable offerings in school lunches are likely to be accepted by children has important implications for school meal policies and children’s health” [5]. On the other side, “eggs and poultry, the researchers found, were the least-wasted food category” [5]. According to our findings, milk products are the most discarded items in both Table II and Table III. FCPS should not worry that students are avoiding their fruits and vegetables, though cranberry products are among the top 10 breakfast items wasted (Table II). Students eating in FCPS schools are eating relatively healthy. However, with milk items frequently being wasted, this points to a problem with their waste management.

TABLE II
Top 10 Most Wasted Breakfast Products

| Product Name | Leftover Rate (%) |
|---|---|
| Fat Free Chocolate Milk(Each) | 98.8 |
| Cereal, Cinnamon Toasters, Malt-O-Meal, IW | 60.3 |
| Fat Free White Milk(Each) | 53.5 |
| Cherry Flavored Dried Cranberries(Each) | 53.1 |
| Fat Free Unflavored Shelf-Stable Milk(Each) | 52.7 |
| Milk, Lactose Free, Fat Free, Shelf-Stable(Each) | 51.3 |
| Orange Flavored Dried Cranberries(Each) | 50.0 |
| Watermelon Flavored Dried Cranberries(2 Eaches) | 48.0 |
| Assorted Cereal(Cereal Cup) | 47.1 |
| Patty, Chicken, Breaded, WG(2 strips) | 46.9 |

TABLE III
Top 10 Most Wasted Lunch Products

| Product Name | Leftover Rate (%) |
|---|---|
| Fat Free White Milk(Each) | 60.6 |
| Milk, Lactose Free, Fat Free, Shelf-Stable(Each) | 50.9 |
| 1% White Milk(Each) | 50.8 |
| Fat Free Unflavored Shelf-Stable Milk(Each) | 48.8 |
| Turkey, Breast, Pre-Sliced, Oven Roasted(6 Slices) | 47.1 |
| String Cheese(Each) | 42.8 |
| 1% Unflavored Shelf-Stable Milk(Each) | 39.9 |
| Assorted 4 oz Yogurt(Container) | 39.0 |
| Mustard Packet(Packet) | 34.7 |
| Lactose Free, White Milk(Each) | 32.9 |

TABLE IV
NEW SCHOOL MENU ITEMS FOR THE CURRENT SCHOOL YEAR

| New Product Name | Category |
|---|---|
| Cinnamon Apple Overnight Oats | Breakfast |
| Egg & Cheese Breakfast Quesadilla | Breakfast |
| Egg & Cheese Breakfast Taco | Breakfast |
| French Toast Sticks | Breakfast |
| Egg & Cheese Bagel | Breakfast |
| Turkey Sausage & Cheese Bagel | Breakfast |
| Sofritas | Lunch |
| Peruvian Chicken Drumstick | Lunch |
| Spanish Rice | Lunch |
| Cheesy Turkey Pepperoni Pasta | Lunch |
| Cheesy Pasta with Marinara Sauce | Lunch |
| Tomato Grilled Cheese | Lunch |
| Deep Dish Pizza | Lunch |
| Chicken Masala Pizza | Lunch |
| Macaroni & Cheese | Lunch |
| Brunch for Lunch (Cheesy Scrambled Eggs, Potato Wedges, French Toast Sticks, Turkey Sausage) | Lunch |
| Veggie Chili | Lunch |
| Patty Melt | Lunch |
| Buffalo Chicken Dip | Lunch |
| Teriyaki Chicken or Chickenless Bites Stir-Fry with Fried Rice and Asian Stir-Fry Vegetables | Lunch |

In their study of the private school in Columbia, Missouri, Mulhollem found that changing the cafeteria menu can be difficult [5]. “Addressing menu changes while staying within budget and offering children nutritious meals often is not an easy task” [5]. In the 2025 school year, Fairfax County Public Schools (FCPS) introduced new menu items [18], as shown in Table IV. Since students may not be familiar with these options, there is a real risk of increased food waste if they choose not to eat them. This could worsen current waste management challenges, especially given the high production costs and large quantities involved, which may result in financial losses. Because these items are new, it will be important to conduct a follow-up study to assess their effectiveness and their impact on waste management.

Another problem brought up is the reliance on a single individual to sustain a waste management program. Huber’s article [4] describes how one teacher, Gerin Hennebaul, initiated the push for change in waste management with her students, which subsequently led to waste-management initiatives. However, if she were the only advocate for this cause and left the school, the program could collapse. In Fairfax County Public Schools, the administration is actively seeking assistance to reduce waste. This approach would require a collective effort from administrative staff and encourage passionate individuals within schools to participate. Though FCPS may be less vulnerable to this issue, it is important that the administration is committed to promoting waste management across the entire school system to ensure widespread adoption.

### B. Limitations

A limitation encountered by other researchers was the narrow scope of the data. In Mulhollem’s article [5], the study observed that this school’s waste percentages align with findings from other U.S. studies. However, these results cannot be generalized to all U.S. public school cafeterias. For our study, we lacked multi-month datasets for analysis. We received food production data only for May. This snapshot did not capture the full extent of the school year’s food production and waste. Consequently, we analyzed May and inferred that the rest of the school year is comparable. Again, this inference does not represent all U.S. public school cafeterias.

We faced uncertainty about how much of the county budget goes to breakfast and lunch production. The budget documents only list unspecified ’Expenditures.’ We assumed the budget includes salaries, maintenance, equipment, utilities, and compliance costs. Therefore, the amounts we present might not fit FCPS if most of a school’s budget covers food production. We also assumed school budgets are allocated by size, not equally. These may not reflect the true amounts each school receives. Our analysis assumes that the Food and Nutritional Services budget primarily funds food production.

## VI. CONCLUSION

This study tackled economic and environmental food waste issues in Fairfax County Public Schools using data analysis and an integer linear programming optimization model. The model projected an annual system-wide cost reduction of about \$1.83 million, lowering spending from \$48.94 million to \$47.11 million. This shows the system can reclaim significant economic value without lowering nutritional standards. The graphs show that the impact of optimization varies by school size. Schools with 500–1500 students had the greatest total savings, while larger schools faced a projected cost increase. This increase does not indicate inefficiency; it reveals historical under-production. The model corrects this to ensure resources fully meet student demand.

This study has some limitations related to the data used and the clarity of the budget. The analysis used production data from May to estimate yearly trends, which might not reflect seasonal changes during a full school year. Another limitation brought up was that the exact use of funds in the ”Expenditures” budget category is unclear; future research will need more detailed financial data to improve cost-saving estimates. Even with these challenges, the study offers a framework for FCPS to move from its current decision-making methods to a more data-driven approach to sustainability. Therefore, FCPS should follow the recommendations generated by our model to achieve their goals to reduce their production of waste from their schools.

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